DECODO Site Unblocker - Hybrid Authentication Guide

Overview

This guide describes the updated DECODO Site Unblocker integration with **hybrid authentication** support:
- **Username:Password** for proxy authentication (unblock.decodo.com:60000)
- **API Key** for additional functions (statistics, management, enhanced features)

Authentication Architecture

1. Proxy Authentication

- **Endpoint**: `unblock.decodo.com:60000`
- **Method**: HTTP Basic Authentication via `Proxy-Authorization` header
- **Format**: `Basic base64(username:password)`
- **Usage**: Required for all proxy requests

2. API Key Authentication

- **Header**: `X-SU-API-Key: your_api_key`
- **Usage**: Optional, enables additional features:
- Usage statistics
- Session management
- Advanced geo-targeting
- Priority routing
- Enhanced support

Configuration

1. Setting Hybrid Credentials

```
// With API Key (recommended)
DECODOConfiguration.shared.setCredentials(
    username: "your_username",
    password: "your_password",
    apiKey: "your_api_key"
)

// Without API Key (basic functionality)
DECODOConfiguration.shared.setCredentials(
    username: "your_username",
    password: "your_password"
)
```

2. Environment Variables

```
# Required for proxy authentication
export DECODO_USERNAME="your_username"
export DECODO_PASSWORD="your_password"

# Optional for enhanced features
export DECODO_API_KEY="your_api_key"
```

Site Unblocker Headers

Basic Headers (Username:Password only)

```
let headers = DECODOConfiguration.createSiteUnblockerHeaders(
    headlessMode: .png,
    geoLocation: "US",
    sessionId: "session_123"
)

// Generated headers:
// X-SU-Headless: png
// X-SU-Geo: US
// X-SU-Session-Id: session_123
// Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8
// ... (standard browser headers)
```

Enhanced Headers (With API Key)

```
let headers = DECODOConfiguration.createSiteUnblockerHeaders(
    headlessMode: .png,
    geoLocation: "US",
    sessionId: "session_123"
)

// Generated headers include:
// X-SU-API-Key: your_api_key  ← Additional API key header
// X-SU-Headless: png
// X-SU-Geo: US
// X-SU-Session-Id: session_123
// ... (all standard headers)
```

Web Scraping API Payload

Basic Payload (Username:Password only)

```
let payload = DECODOConfiguration.createWebScrapingAPIPayload(
    url: "https://example.com",
    target: "universal",
    headlessMode: .html
)

// Generated payload:
// {
//   "url": "https://example.com",
//   "target": "universal",
//   "headless": "html",
//   "device_type": "desktop",
//   "http_method": "GET"
// }
```

Enhanced Payload (With API Key)

```
let payload = DECODOConfiguration.createWebScrapingAPIPayload(
    url: "https://example.com",
    target: "universal",
    headlessMode: .html
)

// Generated payload includes:
// {
//   "api_key": "your_api_key",  ← Additional API key field
//   "url": "https://example.com",
//   "target": "universal",
//   "headless": "html",
//   "device_type": "desktop",
//   "http_method": "GET"
// }
```

Proxy Configuration

Site Unblocker Proxy Setup

```
// Enable Site Unblocker proxy
await ProxyManager.shared.enableDECODOSiteUnblocker()

// The proxy will be configured as:
// - Host: unblock.decodo.com
// - Port: 60000
// - Authentication: username:password via Proxy-Authorization
// - Additional headers: X-SU-API-Key (if available)
```

Request Flow

```
1. Client Request
   ↓
2. Proxy-Authorization: Basic base64(username:password)
   ↓
3. X-SU-API-Key: your_api_key (if available)
   ↓
4. unblock.decodo.com:60000
   ↓
5. Target Website
```

UI Integration

Settings View Updates

The `DECODOSettingsView` now includes:

1. **Username Field**: Required for proxy authentication
2. **Password Field**: Required for proxy authentication
3. **API Key Field**: Optional for enhanced features
4. **Save Button**: Stores all credentials securely in Keychain
5. **Test Buttons**: Validate both proxy and API functionality

Usage Example

```swift
struct ContentView: View {
    var body: some View {
        NavigationView {
            VStack {
                // ... other content

                NavigationLink("DECODO Settings") {
                    DECODOSettingsView()
                }
            }
        }
    }
}
```

Testing

Unit Tests

```
func testHybridAuthentication() {
    // Test with API key
    DECODOConfiguration.shared.setCredentials(
        username: "test_user",
        password: "test_pass",
        apiKey: "test_api_key"
    )

    let headers = DECODOConfiguration.createSiteUnblockerHeaders()
    XCTAssertEqual(headers["X-SU-API-Key"], "test_api_key")
}
```

Integration Tests

```
func testRealConnection() async {
    guard let username = ProcessInfo.processEnvironment["DECODO_USERNAME"],
          let password = ProcessInfo.processEnvironment["DECODO_PASSWORD"] else {
        throw XCTSkip("Real credentials not available")
    }

    let apiKey = ProcessInfo.processEnvironment["DECODO_API_KEY"]

    DECODOConfiguration.shared.setCredentials(
        username: username,
        password: password,
        apiKey: apiKey
    )

    let result = await DECODOConfiguration.shared.testSiteUnblockerConnection()
    XCTAssertTrue(result.success)
}
```

API Key Benefits

When an API key is provided, you get access to:

1. Enhanced Statistics

- Request/response metrics
- Bandwidth usage tracking
- Success/failure rates
- Geographic distribution

2. Session Management

- Persistent sessions across requests
- Session-based IP rotation
- Custom session timeouts
- Session state tracking

3. Priority Support

- Faster response times
- Priority routing
- Dedicated support channels
- Advanced troubleshooting

4. Advanced Features

- Custom geo-targeting options
- Enhanced captcha solving
- Premium proxy pools
- Advanced retry logic

Error Handling

Common Issues

1. **“Proxy Authentication Required”**
   - Verify username/password are correct
   - Check credentials in DECODO dashboard
   - Ensure proxy endpoint is accessible

2. **“API Key Invalid”**
   - Verify API key format
   - Check API key permissions
   - Regenerate API key if needed

3. **“Connection Timeout”**
   - Check network connectivity
   - Verify proxy endpoint availability
   - Try different geo-location

Debug Information

```
func debugDECODOStatus() {
    let config = DECODOConfiguration.shared
    print("DECODO Status:")
    print("- Credentials configured: \(config.hasValidCredentials())")
    print("- Connection status: \(config.connectionStatus)")
    print("- API key available: \(config.credentials?.apiKey != nil)")

    if let lastTest = config.lastTestResult {
        print("- Last test: \(lastTest.success ? "Success" : "Failed")")
        print("- Latency: \(lastTest.formattedLatency)")
        if let error = lastTest.error {
            print("- Error: \(error)")
        }
    }
}
```

Migration from Previous Version

Before (Username:Password only)

```
DECODOConfiguration.setCredentials(username: "user", password: "pass")
let headers = DECODOConfiguration.createSiteUnblockerHeaders()
// No API key functionality
```

After (Hybrid Authentication)

```
// Backward compatible - existing code works
DECODOConfiguration.shared.setCredentials(username: "user", password: "pass")

// Enhanced - with API key
DECODOConfiguration.shared.setCredentials(
    username: "user",
    password: "pass",
    apiKey: "api_key"
)

let headers = DECODOConfiguration.createSiteUnblockerHeaders()
// Automatically includes X-SU-API-Key if available
```

Best Practices

1. Security

- Store credentials in Keychain (automatically handled)
- Never hardcode credentials in source code
- Use environment variables for testing
- Rotate API keys regularly

2. Performance

- Reuse sessions when possible
- Use appropriate geo-targeting
- Monitor usage statistics
- Implement proper retry logic

3. Error Handling

- Always check connection test results
- Implement fallback mechanisms
- Log errors for debugging
- Provide user feedback

Support

Getting Credentials

1. Sign up at decodo.com (https://decodo.com)
2. Access your dashboard
3. Generate username/password for proxy

4. Generate API key for enhanced features
5. Configure in app settings

Troubleshooting

1. Test connection in app settings
2. Check DECODO service status
3. Verify credentials in dashboard
4. Contact DECODO support if needed

Resources

- DECODO Website (https://decodo.com)
- DECODO Dashboard (https://dashboard.decodo.com)
- Site Unblocker Documentation (https://help.decodo.com/docs/site-unblocker-quick-start)
- API Documentation (https://help.decodo.com/docs/api)

This hybrid authentication system provides full backward compatibility while enabling enhanced features through optional API key integration.